Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

OptiCrop aims to revolutionize agricultural practices by leveraging machine learning to provide intelligent crop recommendations. By analyzing soil nutrients and environmental parameters, the system addresses key challenges in modern farming including poor crop selection, inefficient resource usage, and reduced agricultural productivity. Key features include an ML-based crop recommendation model, real-time prediction through a Flask web application, and data-driven decision support for farmers and agricultural stakeholders.

Project Overview	
Objective	To develop an intelligent ML-based crop recommendation system that analyzes soil and environmental parameters (N, P, K, temperature, humidity, pH, rainfall) and recommends the most suitable crop for maximum agricultural yield and resource efficiency.
Scope	The project covers data collection, preprocessing, ML model training and evaluation, Flask web application development, and local deployment. The system targets farmers, agricultural researchers, agribusiness organizations, and policymakers.
Problem Statement	
Description	Farmers lack access to intelligent, data-driven tools for crop selection. Poor crop decisions based on incomplete knowledge of soil chemistry and climate conditions lead to financial losses, low yields, and inefficient use of resources like fertilizers and water.
Impact	Solving this problem will result in improved crop yields, optimized resource utilization, reduced agricultural losses, and better farming decisions. It will contribute to sustainable farming practices and support food security at regional and national levels.
Proposed Solution	
Approach	Employing supervised and unsupervised machine learning techniques to analyze agricultural datasets containing soil and climate parameters. The system trains multiple ML models, selects the best performer, and integrates it with a Flask web application for real-time crop prediction.
Key Features	- ML-based crop recommendation using N, P, K, temperature, humidity, pH, and rainfall - Multiple ML algorithms: KNN, Logistic Regression, Decision Tree, Random Forest, K-Means Clustering - Real-time prediction through a responsive Flask web interface - Model saved as .pkl file for fast reuse without retraining - Data preprocessing: null handling, outlier treatment, feature scaling - Visual data analysis using Matplotlib and Seaborn

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Processor specifications	Intel Core i3 or above / M2 MacBook
Memory	RAM specifications	Minimum 4 GB RAM (8 GB recommended)
Storage	Disk space for data, models and logs	Minimum 10 GB free storage
Software		
Frameworks	Python web framework	Flask
Libraries	Additional ML libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	IDE / Notebook	Jupyter Notebook, PyCharm, VS Code
Version Control	Repository management	Git + GitHub
Data		
Dataset	Source, size, format	Kaggle Agricultural Dataset — Crop Recommendation Dataset 2200 records, 8 features (N, P, K, temperature, humidity, pH, rainfall, label), CSV format